Chapter: 24

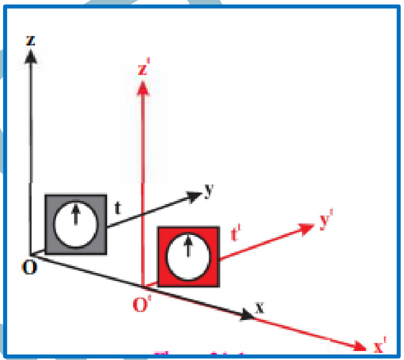
RELATIVITY

FRAME OF REFERENCES:

Definition:

> *"A rigid framework (usually x, y and z-axes) relative to which the position and motion of an object can be measured is called frame of references". OR*
>
> *"Any coordinate system, relative to which measurements are taken is called frame of reference".*

Explanation:



A reference frame is a fundamental concept that helps us to describe and understand the relative motion of objects and events.

The simplest frame of reference is the Cartesian coordinate system which consists of three mutually perpendicular lines i.e. x-axis, y-axis and z-axis.

> *It consists of a set of coordinate axes (usually x, y, and z axes) and a clock, allowing observers to precisely define where and when events occur.*

The familiar Cartesian System of Coordinates, in which the position of the particle is specified by its three co-ordinates x, y, z, along the three perpendicular axes, is shown in figure.

We have indicated two observers O and O' and a particle P. These observers use frames of reference XYZ and X'Y'Z' respectively.

- If O and O' are at rest, they will observe the same motion of P.
- But if O and O' are in relative motion, their observation of the motion of P will be different.

Examples:

1. The position of table in a room can be located (or described) relative to the walls of the room. The room is then the frame of reference.
2. In order to make the measurement in the laboratory, the laboratory is the frame of reference.
3. If the same experiment is performed on the train, the train becomes the frame of reference.

TYPES OF FRAMES OF REFERENCE:

There are two types of frames of reference.

1. Inertial frame of reference. 2. non-inertial frame of reference.

1. INERTIAL FRAME OF REFERENCE:

Definition:

> *"A coordinate system in which the law of inertia is valid, is called Inertial frame of reference." OR*
>
> *"A frame of reference moving with uniform velocity with respect to the other frame of reference or at rest is called an inertial frame of reference."*

It can also be said as, non-accelerated frame of reference.

Explanation:

If we place a body upon the Earth, it remains at rest unless an unbalanced force is applied to it. This observation shows that Earth is considered as an inertial frame of reference.

If a body placed in a car moving with uniform velocity with respect to earth remains at rest, then a moving car is also an inertial frame of reference. Thus, any frame of reference which is moving with uniform velocity relative to an inertial frame is also an inertial frame.

2. NON-INERTIAL FRAME OF REFERENCE:

Definition:

> *"A coordinate system in which law of inertia is not valid is called non-inertial frame of reference." OR*
>
> *"If a frame of reference is moving with certain acceleration with reference to another frame of reference, this is called an accelerated frame of reference or non-inertial frame of reference.*

Explanation:

If a moving car is suddenly stopped, the body placed in it will not remain at rest because the car is suddenly retarded (negative acceleration). So is the case when the car suddenly accelerated. In such a situation, the car is not an inertial frame of reference. Hence, an accelerated frame is a non-inertial frame of reference.

Earth is rotating and revolving and hence the Earth is not an inertial frame but non inertial due to its spinning or orbital motion. But it can often be treated as an inertial frame without serious error due to very small value of acceleration.

DIFFERENCE:

Criteria	Inertial frame of Reference	Non-Inertial frame of Reference
Definition	A frame of reference with a constant velocity.	A frame of reference with an accelerating motion.
Motion of Objects	Objects in uniform motion appear to follow straight lines or constant velocities.	Objects may appear to accelerate or experience fictitious forces.
Newton's first law	Newton's first law (Law of inertia) is valid in this frame.	Newton's first law is not valid due to accelerating motion.
Appearance of forces	Real forces are observed and can be directly measured.	Fictitious forces (e.g., centrifugal force, Coriolis force) may appear due to the acceleration of the frame.
Equations of motion	Newton's laws of motion hold true in this frame.	Additional terms or transformations may be required to account of the frame
Examples	A person inside a moving train, an object in free fall.	A person inside a spinning carousel, an object in circular motion
Application	Often used in analyzing motion and dynamics in	Important in understanding phenomena in rotating systems, general relativity, and

RELATIVE MOTION:

Relative motion is a concept that describes the motion of an object in relation to another observer or frame of reference.

This viewpoint recognizes that motion is not absolute but depends on the observer's point of view.

1. WHEN TWO OBJECTS ARE MOVING IN THE SAME DIRECTION:

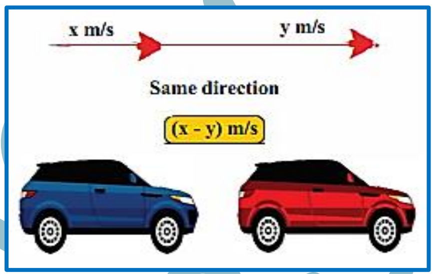
When two objects are moving in the same direction, their relative motion involves the difference in their velocities.

The relative velocity is the velocity of one object as seen from the perspective of the other.

The formula for relative velocity (V_{rel}) when the objects move in the same direction is given by:

$$V_{rel} = V_1 - V_2$$

where V_1 and V_2 are the velocities of the two objects, respectively.



Example:

Consider two cars, A and B, moving on a straight road. If car A has a velocity of 20 m/s, and car B has a velocity of 15 m/s in the same direction, the relative velocity of car B as observed from car A is

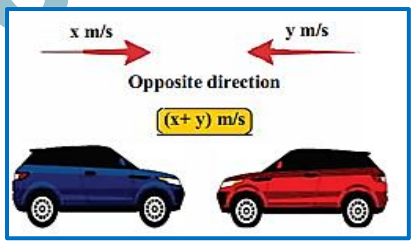
$$V_{rel} = 20m/s - 15m/s = 5m/s$$

2. WHEN TWO OBJECTS ARE MOVING IN THE OPPOSITE DIRECTION:

When two objects are moving in opposite directions, their relative motion involves the sum of their velocities.

The formula for relative velocity (V_{rel}) when the objects move in opposite directions is given by:

$$V_{rel} = V_1 + V_2$$



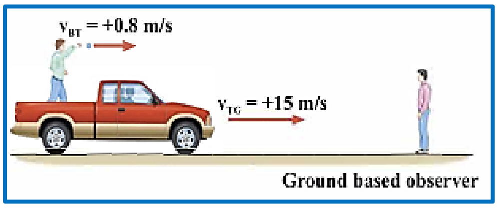
Example:

Consider the same two cars, A and B. But this time moving in opposite directions. Let's say car A is traveling at 80 m/s, and car B is now approaching it at 50 m/s. For a passenger in car A, car B appears to be moving towards them at relative velocity:

V_{rel} = 80 m/s + 50 m/s= 130 m/s

FRAMES OF REFERENCE: PREDICTION OF MOTION:

Consider a vehicle moving at a constant speed v = 15 m/s with relative to the ground and a boy standing on the vehicle throws a ball with velocity v = 8 m/s in the same direction as the vehicle's motion. As shown in figure.



If someone on the ground measures the ball's speed, it intuitively seems like it should be 23 m/s (15+8).

Conversely, if you throw the ball in the opposite direction of the vehicle's speed, the ground observer would measure a speed of 7 m/s (15 - 8).

Both observers are measuring the speed of the same object, but their relative motion leads to different results.

Derivation of general equation for relative Motion:

Now, let's derive a general equation applicable to any speeds. In our example, the vehicle's speed of 15 m/s becomes the relative speed of the two reference frames, denoted as u. The speed of the thrown ball is v' (in the train's frame), and the measured speed on the ground is v.

In a more formal way, we can express this relationship as:

$$v = v' + u$$

This equation is essentially the **Galilean transformation** of speeds. But it is only applicable in mechanics.

GALILEAN TRANSFORMATION EQUATIONS:

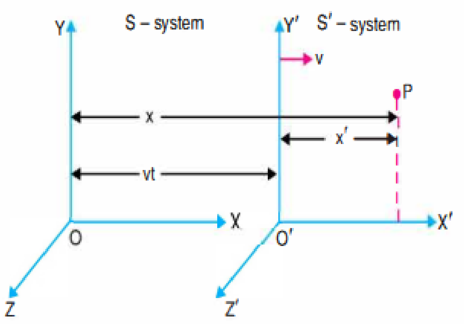
"The equations, describe the relationship between the coordinates of an event in two different inertial frames of references are called Galilean transformation equations".

Explanation:

These equations were formulated by Galileo and are applicable when the relative velocities involved are much smaller than the speed of light. With these equations, we can correctly measure the velocities of a ball in the above example.

Galileo concluded that:

"The laws of mechanics must be valid in all inertial frames of reference."



This is also known as the "Principle of Galilean Relativity".

Let x and x' be two inertial frames (Figure). Let x be at rest and x' moves with uniform velocity "v" along the positive X direction.

We assume that: v << c.

Let the origins of the two frames coincide at t = 0.

Suppose some event occurs at the point P. The observer O in frame x determines the position of the event by the coordinates x, y, z. The observer O' in frame x' determines the position of the event by the coordinate's x', y', z'.

Let the time elapsed at the same rate in both frames, i.e., t = t'. There is no relative motion between x and x' along the axes of Y and Z.

Measurements in the X direction made x frame will be greater than those made in x' frame by the amount vt, which is the distance x' has moved in the X direction.

Therefore, $x' = x - vt$

$y' = y$; $z' = z$ and $t' = t$

The set of the above equations is known as the Galilean transformation equations.

PRINCIPLE OF RELATIVITY:

RELATIVITY:

Relativity, a fundamental theory in physics developed by Albert Einstein, revolutionized our understanding of space, time, and energy.

Albert Einstein assumed that:

All possible reference frames moving at uniform velocities relative to one another (i.e. inertial frames) are equivalent for the statement and description of physical laws.

This simple assumption is called the principle of Relativity, based on the constancy of the velocity of light Einstein further generalized this principle to restate it in the form.

"All inertial frame of references are completely equivalent for all physical phenomena"

THEORIES ABOUT RELATIVITY:

There are two main theories about relativity:

1. Special Theory of Relativity.
2. General Theory of Relativity.

1. SPECIAL THEORY OF RELATIVITY:

"The theory related to the relative motion of inertial frames of reference, is called special theory of relativity."

Special Relativity, and the idea that time and space are relative, vary according to the observer's motion.

2. GENERAL THEORY OF RELATIVITY:

"The theory related to the relative motion of inertial as well as non-inertial frames of reference, is called general theory of relatively."

General Relativity introduced extends these ideas to include gravity, describing it not as a force but as the curvature of space time caused by mass and energy.

1. SPECIAL THEORY OF RELATIVITY:

Definition:

"The theory related to the relative motion of inertial frames of reference, is called special theory of relativity."

Explanation:

The theory of relativity is concerned with the way in which observers who are in a state of relative motion describe physical phenomena.

The special theory of relativity treats problems involving inertial or non-accelerating frames of reference.

The special theory of relativity explains how to interpret motion between different inertial frames of reference.

In other words:

It deals with the problems in which one frame of reference moves with a constant linear velocity relative to another frame of reference.

POSTULATES OF SPECIAL RELATIVITY:

The special theory of relativity is based on two essential assumptions, commonly known as postulates.

POSTULATE: I (PRINCIPLE OF RELATIVITY):

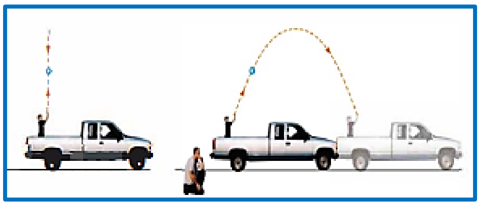
The laws of Physics have the same form in all inertial frames of reference.

Explanation:

The first postulate is the generalization of the fact that all physical laws are the same in frames of reference moving with uniform velocity with respect to one another.

If the laws of physics were different for different observers in relative motion, the observer could determine from this difference that, which of them were stationary in a space and which were moving.

But such a distinction does not exist, so this postulate implies that there is no way to detect absolute uniform motion.



Example:

Consider a vehicle moving at a constant speed. Inside the vehicle, a passenger throws a ball straight up into the air. If we ignore any effects from the air, the passenger inside the moving vehicle sees the ball move up and then back down in a straight line, just as it would appear if someone standing still on the Earth threw a ball upwards.

This means that the laws of physics that govern the motion of the ball, including gravity and equations for constant acceleration, work the same way whether the vehicle is moving or at rest.

Both observers, the one inside the vehicle and the one on the ground, see the ball go up and come back down.

However, they see the path of the ball differently. The person on the ground sees the ball's path as a curved shape (a parabola), while the person in the vehicle sees it as a simple up-and-down motion.

Additionally, the person on the ground thinks that the ball has a horizontal component of velocity, which is the same as the vehicle's velocity. For the outside observer, the distance traveled along the parabolic path is longer than the path straight down but the time for the fall is the same. For the outside observer, the average velocity is greater.

POSTULATE: II (PRINCILPE OF CONSTANCY OF THE SPEED OF LIGHT):

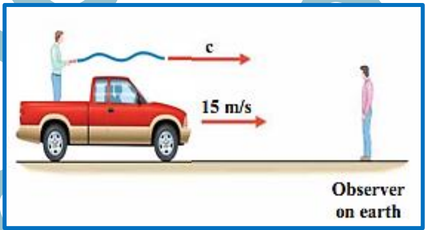
> *The speed of light in vacuum has the same value ($c = 3 \times 10^8$ m/s) in all inertial reference frames, regardless of the velocity of the observer or the velocity of the source emitting the light.*

Explanation:

The second postulate states an experimental fact that speed of light in free space is the universal constant "c" ($c = 3 \times 10^8$ m/s).

Example:

Consider again the same example of throwing a ball in a moving vehicle (as shown in figure) but this time, instead of throwing a ball, we shine a flashlight. If we were to do the same thing with light as we did with the ball, common sense might suggest that the speed of light would increase if the vehicle were moving in the same direction as the light.



According to Galilean Transformation Equation:

$v = c + u$ (If we replace ball's speed v' with speed of light c).

In fact, such an increase in the speed of light has never been found.

In experiments carried out to test the effect of the movement of the source on the speed of light (Michelson-Morley), the results indicate that the speed of light is completely unaffected by the motion of the source. It appears that the speed of light in a vacuum is constant regardless of relative motion.

LORENTZ TRANSFORMATION EQUATIONS:

The Lorentz transformation equations are fundamental in the theory of Special Relativity, describing how measurements of space and time change for observers moving relative to each other at constant velocities.

In order to correct the Galilean equation of velocity addition, we have to consider velocity of light too. This correction was done by Hendrik Lorentz. The resultant equations are known as Lorentz transformation equations.

RELATIVE MOTION: LORENTZ TRANSFORMATION EQUATIONS:

For an event with space-time coordinates (x, y, z, t) in one inertial frame (say, frame S), and (x', y', z', t') in another inertial frame (frame S') moving at a constant velocity v relative to S along the x-axis, the Lorentz transformation equations are

$$x' = \gamma (x - v\,t); \quad y' = y; \quad z' = z; \quad t' = \gamma \left(t - \frac{v.x}{c^2}\right)$$

Where c is the speed of light, and $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ is the Lorentz factor.

SELF-ASSESSMENT QUESTIONS:

1. Calculate the value of the Lorentz factor if the speed of the car is 20m/s. Can we use Galilean Transformation equation for this value?
2. Why can't objects with mass e½ceed the speed of light?

CONSEQUENCES OR RESULTS OF SPECIAL THEORY OF RELATIVITY:

In Relativity, there is no such thing as an absolute length or absolute time interval.

Here are some of the most important consequences of special relativity:

1. RELATIVITY OF SIMULTANEITY:

The principle of relativity states that there is no preferred inertial frame of reference.

Definition:

Two events that are simultaneous in one reference frame are in general but not simultaneous in a second frame moving relative to the first.

Simultaneity depends on the state of motion of the observer and is therefore not an absolute concept.

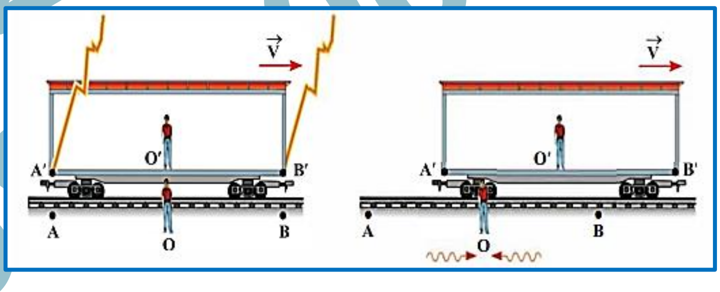
Explanation:

The relativity of simultaneity is a concept in Albert Einstein's theory of relativity. It suggests that two events that occur simultaneously in one frame of reference may not be simultaneous in another frame of reference that is moving relative to the first.

Essentially, simultaneity is relative depending on the observer's motion. This has profound implications for our understanding of time and space.

Example:

Imagine two observers, Boy A is standing on a train platform and boy B is on a moving train. They are equidistant from the center O of the platform. At the exact moment when the train passes from the center of the platform, lightning strikes both ends of the train. As shown in figure.



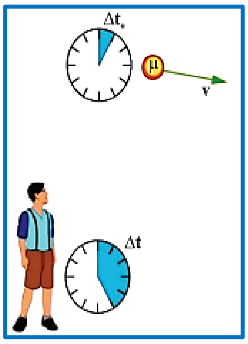
1. **From boy A's perspective:** Boy A sees the lightning strikes happen at the same time because he is stationary relative to the platform. Since light travels at a finite speed, the light from both lightning strikes reaches boy A simultaneously.
2. **From boy B's perspective:** Boy B is sitting on the moving train. Because the train is moving toward the lightning strike at the front of the train and away from the lightning strike at the back of the train, light takes longer to reach him from the back of the train than from the front.

   As a result, Boy B perceives the lightning strike at the front of the train before the lightning strike at the back of the train. Thus, for boy B, the lightning strikes are not simultaneous.

This example demonstrates how the perception of simultaneity can differ between observers depending on their relative motion. In this case, what appears simultaneous to boy A (the lightning strikes) does not appear simultaneous to boy B due to his motion relative to the events.

2. TIME DILATION:

The time interval between two events occurring at a given point in the moving frame (S') appears to be longer to the observer in the stationary frame (S). This effect is called time dilation.



Explanation:

Since, in classical physics time is a constant quantity, but according to the special theory of relativity, time is not constant or absolute quantity. It depends upon the motion of the frame of reference.

Let Δt_o be the proper time measured by a clock that is at rest. The relative time "t" measured in another frame of reference is given by: $\Delta t = \dfrac{\Delta t_o}{\sqrt{1-\frac{v^2}{c^2}}}$

Where Δt is the relativistic time and "c" is speed of light (3×10^8m/s).

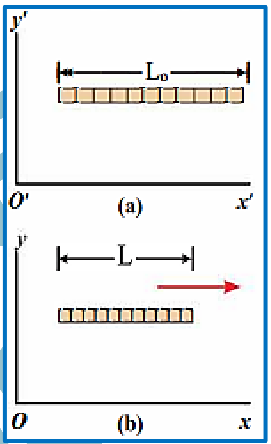
Here, the value of $\sqrt{1-\frac{v^2}{c^2}}$ should not be equal to zero; this will only occur when the speed of an object is less than the speed of light c.

If the speed of an object "v" equals "c" in the above equation, then Δt becomes infinite.

This implies that time will seemingly stop, which is impossible. Therefore, no material object can travel at the speed of light.

3. LENGTH CONTRACTION:

"When the velocity of an object increases, the length of the object decreases and at the velocity of light become zero."



Explanation:

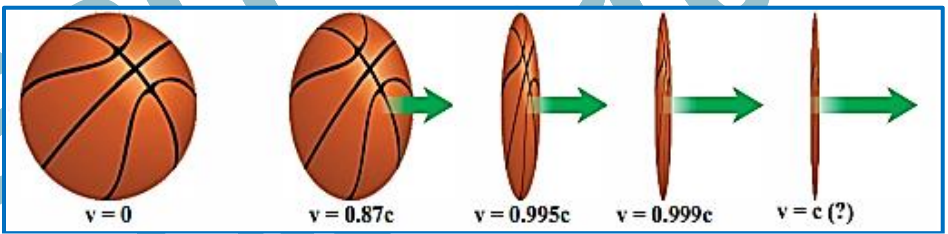
Let "L_o" be the length of a rod when the rod is stationary as shown in figure (a). If there is relative motion at speed "v" between an observer at rest and the rod along the length "L" of the rod, then observer will calculate a relativistic length L given by

$$L = L_o\sqrt{1-\frac{v^2}{c^2}}$$

Where the length L_o is called proper length, L is relativistic length and c is speed of light (3×10^8 m/s).

Relative motion causes a length contraction as shown in figure (b).

A greater speed "v" results in greater contraction. Space contracts in only one direction, the direction of motion as shown in figure.



4. MASS VARIATION: INCREASE IN MASS

"When the velocity of an object increases, the mass of the object increases and at the velocity of light mass become infinite."

Explanation:

According to the special theory of relatively, the mass of an object in a frame of reference at rest, is called its rest mass "m_0". If this mass is measured by an observer moving with a constant speed "v" relative to the object, then the mass "m" in the moving frame will increase.

Let "m_o" is the rest mass of an object. If the object is moving at speed "v" then its relativistic mass m will be given by

$$m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$$

Where m is relativistic mass and c is speed of light. Relative motion causes a mass variation

5. MASS ENERGY RELATIONSHIP:

"Mass and energy are interconvertible."

Explanation:

This conclusion is quite different than classical physics which had stated the laws of conservation of mass and energy, separately.

According to Einstein's special theory of relativity, mass and energy are interchangeable. An object's mass m_o and the equivalent energy E_o are related by:

$$E_o = m_o c^2$$

Where "c" is the speed of light. This equation gives the rest mass energy E_O, that is associated with the object's mass m_o, regardless of whether the object is at rest or moving.

If the object is moving, it has additional energy in the form of kinetic energy K. The total energy E is the sum of its rest mass energy and its kinetic energy:

$$\begin{aligned} E &= E_o + K \\ &= m_o c^2 + K \\ E &= \gamma m_o c^2 \end{aligned}$$

Where γ is the Lorentz factor for the object's motion. And E is the total relativistic energy of the object.

From the above equations, relative kinetic energy can be calculated as:

$$\begin{aligned} K &= E - E_o \\ &= E - m_o c^2 \\ &= \gamma m_o c^2 - m_o c^2 \\ K &= m_o c^2 (\gamma - 1) \end{aligned}$$

MAXIMUM VELOCITY OF A PARTICLE:

A basic result of the special theory of relativity is that:

The speed of an object cannot equal or exceed the speed of light.

That the speed of light is a natural speed limit in the universe can be seen from the equations of mass variation, length contraction, time dilation and mass-energy relationship.

INCREASE IN MASS:

If an object is accelerated to greater speeds, its mass becomes larger and larger. Indeed, if "v" is equal to c, the denominator in the Lorentz equation would be zero and the mass would become infinite.

To accelerate an object up to v = c would thus require infinite energy, and so it is not possible.

TIME DILATION:

Special relativity also predicts time dilation, meaning that as an object approaches the speed of light, time appears to slow down for the object relative to an observer at rest. This effect becomes more pronounced as the object's velocity approaches "c".

Consequently, from the perspective of an observer, it would take an infinite amount of time for a massive object to reach the speed of light.

LENGTH CONTRACTION:

Another consequence of special relativity is length contraction, where objects moving at relativistic speeds appear shorter in the direction of motion when observed from a stationary frame.

This effect prevents objects from achieving relativistic velocities within a finite distance because, as the object's velocity increases, its length contracts in the direction of motion, making it increasingly difficult to accelerate further.

RELATIVISTIC EFFECTS ON SPACE TRAVEL:

The implications of mass increase, time dilation, and length contraction for space travel are profound and have significant consequences for our ability to explore the universe at relativistic speeds.

Here's how each of these effects impacts space travel:

1. MASS INCREASE:

- The implication for space travel is that as spacecraft approach relativistic speeds, their mass increases exponentially. This makes it increasingly difficult to accelerate them further, requiring enormous amounts of energy.
- Overcoming the mass increase becomes a significant engineering challenge for spacecraft propulsion systems. Current propulsion technologies, such as chemical rockets, would become impractical at relativistic speeds due to the massive energy requirements.

2. TIME DILATION:

- For space travelers moving at relativistic speeds, time dilation means that their perception of time differs from that of stationary observers. A journey that may take several years according to Earth-based observers could be experienced as much shorter by the travelers due to time dilation.
- Time dilation has implications for interstellar travel, where long-duration missions could be undertaken without experiencing the full effects of time passing, as perceived by Earth-based observers. However, it also presents challenges for communication and synchronization with mission control on Earth.

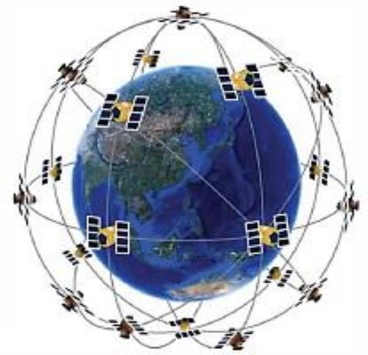
3. LENGTH CONTRACTION:

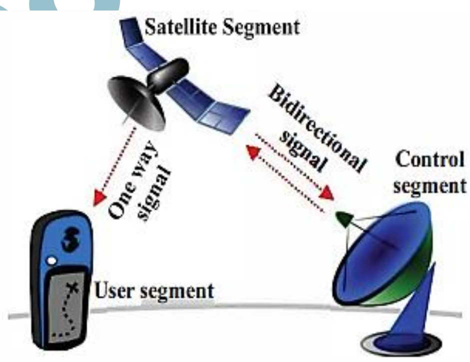
- For space travelers, length contraction means that distances along their direction of motion appear shorter. This has implications for navigation and spacecraft design, as distances may be perceived differently by travelers moving at relativistic speeds compared to stationary observers.
- Additionally, length contraction affects the perception of space travel distances. What may appear as a vast distance to stationary observers could be contracted from the perspective of travelers moving at relativistic speeds.

NAVIGATION SATELLITE TIMING AND RANGING (NAVSTAR) SYSTEM:

It is the official name for the Global Positioning System (GPS).

It's a network of 24 satellites orbiting Earth that allows us to determine our location on the planet with incredible accuracy.





One of the core principles of NAVSTAR system is synchronized clocks on satellites. These clocks tick incredibly accurately, but due to the effects of relativity, this isn't as simple as it seems. Here's why:

1. TIME DILATION:

As the satellites zoom around Earth at high speeds (roughly 11,000 km/h), their clocks actually run slightly slower compared to clocks on Earth.

2. GRAVITY:

The satellites being closer to Earth experience a stronger gravitational pull, which, according to General Relativity, also slows down their clocks compared to those on the ground.

If these effects weren't accounted for, our GPS calculations would be significantly off. But, because scientists understand space-time and its relativistic effects, they can account for this time discrepancy and maintain the accuracy of GPS positioning.

Using Equation of time dilation and the speed of the GPS satellites, we can calculate the difference between the dilated time interval and the proper time interval as a fraction of the proper time interval and compare the result to the stability of the GPS clocks:

$$\frac{\Delta t - \Delta t_o}{\Delta t_o} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1$$

$$= \frac{1}{\sqrt{1 - (4000)^2/(3 \times 10^8)^2}} - 1$$

$$= \frac{1}{1.1 \times 10^{10}}$$

This indicates the level of stability. It means the clock deviates by one part in 10 trillion (10^{13}) over a certain period.

In other words, the clock's timekeeping is extremely precise, with an error margin of just 0.000000000 l%.

WORKED EXAMPLES:

24.1: The period of a pendulum is measured to be 3.00 sin the inertial frame of the pendulum. What is the period, as measured by an observer moving at a speed of 0.950c with respect to the pendulum?

24.2: A starship is measured to be 125 m long while it is at rest with respect to an observer. If this starship now flies past the observer at a speed of 0.99c, what length will the observer measure for the starship?

24.3: At what speed will an object's relativistic mass be twice its rest mass?

GENERAL THEORY OF RELATIVITY:

Statement:

The General Theory of Relativity states that:

> *"Gravity is not a force acting at a distance; instead, it is the result of the curvature of spacetime produced by the presence of mass and energy."*

Explanation:

The General Theory of Relativity, proposed by Albert Einstein, explains gravity as a result of the curvature of spacetime.

Massive objects bend spacetime, and other objects move along these curved paths, which appear as gravitational attractions.

According to the General Theory of Relativity:

> *Gravity arises due to the curvature of spacetime produced by mass and energy.*

In simple words mass and energy tell spacetime how to curve, and curved spacetime tells objects how to move.

Space and time form a single fabric called spacetime. Massive objects (like planets, stars) **bend spacetime** around them.

Objects move along the curved paths (called geodesics) in spacetime, and this motion is what we observe as gravity.

In Short:

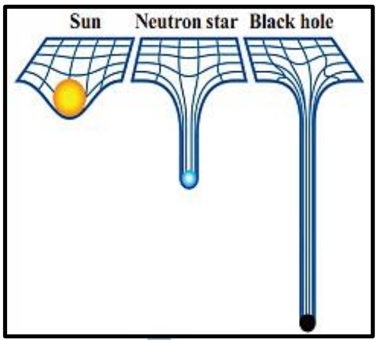
According to General Relativity:

- Space and time together form spacetime
- Heavy objects like the Sun or Earth bend spacetime
- Smaller objects move along this bend
- This bending effect is what we call gravity

Hence, Gravity is not a force, it is an effect of curved spacetime.

SPACE TIME:

Spacetime is a four-dimensional continuum that includes three dimensions of space and one dimension of time.



Explanation:

Einstein combined space (three dimensions) and time into a four-dimensional fabric called "Space-Time"

Imagine it as a two-dimensional rubber sheet. If you place a heavy object, like a bowling ball, on the sheet, it will cause the sheet to deform or bend around the object.

Now, if you roll a smaller ball across the sheet, it will move along the curved paths created by the heavy object.

This is analogous to how planets and other celestial bodies move along the curved space-time around massive objects like the Sun. As illustrated in Figure.

POSTULATES OF GENERAL RELATIVITY:

The general theory of relativity is based on the following two postulates.

Postulate: I. (General Principle of Relativity)

The laws of physics can be expressed in equations that have the same form in all frames of reference, regardless of their state of motion.

Unlike Special Relativity, which applies only to inertial (constant velocity) frames, this postulate extends the laws of physics to cover accelerated motion as well. This extension allows the theory to describe gravitational phenomena.

Postulate: II (Principle of equivalence):

There is no way to spot the difference between gravity and acceleration for an observer in a closed laboratory.

This postulate follows from the experimental observation that the inertial mass of a body is always exactly equal to its gravitational mass.

CONSEQUENCES OF GENERAL THEORY OF RELATIVITY:

General relativity has a number of consequences:

1. Space-Time as a Fourth Dimension:

Einstein's general theory of relativity states that time is a fourth dimension added to the three dimensions of space. Einstein called this four-dimensional geometry, as space-time.

2. Gravitational Red shift:

The general theory of relativity also predicted light coming from a strong gravitational field would have its wavelength shifted toward longer wavelengths. This phenomenon is known as **gravitational red shift.**

3. Existence of Black Holes:

The theory also predicted that when gravity becomes great enough, it would produce objects called black holes.

Black holes are objects whose gravity is so massive that light cannot escape from the surface at all.

Since no light can escape, such objects would appear black.

4. Gravitational Lensing:

Since gravity bends space-time, **light traveling near a massive object will follow a curved path**.

This bending of light can cause distant objects (like stars or galaxies behind a massive object) to appear distorted, magnified, or shifted in position.

This effect is called **Gravitational Lensing** and is used by astronomers to study distant parts of the universe.

5. The expansion of the universe:

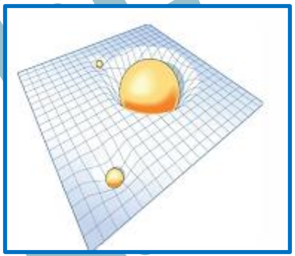
General relativity predicts that the universe is expanding. This has been confirmed by astronomical observations.

6. Gravitational Time Dilation (Practical Application in GPS)

A practical consequence of general relativity is that **gravity slows down time**. Clocks closer to a massive object (where gravity is stronger) tick slower than clocks further away

CONCLUSIONS: GRAVITY AS SPACE TIME CONTINUUM:

The concept of **Gravity as a Space-Time Continuum** replaces the old idea of gravity being a "force." Instead, it describes gravity as a change in the geometry (shape) of the universe itself.



According to General Theory of Relativity:

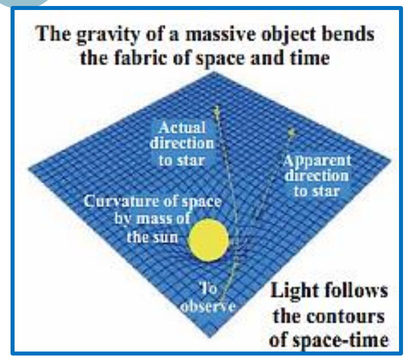
When you put mass in space-time, it bends the geometry of space-time.

According to this theory, gravity is not just a force between masses, as described by Newtonian physics, but rather a manifestation of "c" the curvature of space-time caused by mass and energy.

Here's a breakdown of what this concept entails:

1. SPACE-TIME CONTINUUM:

- Einstein proposed that the three dimensions of space (length, width, height) and the one dimension of time are not separate. They are unified into a single, four-dimensional entity called **space-time**.
- In the absence of heavy objects, this space-time fabric is flat



2. CURVATURE OF SPACE-TIME:

- According to the General Theory of Relativity, when you place a massive object (like a star or planet) in space-time, it **warps or bends** the fabric around it.
- Gravity is not a force reaching out to pull things; rather, it is the **curvature of space-time** caused by mass and energy.
- The heavier the object (greater mass), the greater the curvature.

3. EFFECTS OF GRAVITY:

- Objects in space always try to follow the shortest possible path.
- In flat space, the shortest path is a straight line.
- In curved space-time, the shortest path is a curve. These curved paths are called **geodesics**.

4. PHYSICAL INTERPRETATION:

- Gravity is thus interpreted as the result of objects moving through the curved space-time created by mass and energy.
- Massive objects "warp" the space-time around them, causing other objects to move in response to this curvature.
- This concept provides a unified explanation for both the gravitational force and the motion of objects in space, incorporating both space and time into a single framework.

SUMMARY

- Special Reactivity applies to objects moving at constant velocity relative to each other. It introduces the concept of reference frames and explains how the laws of physics are the same in all inertial reference frames.
- Two important postulates of Special Relativity are:
  1. The laws of physics are the same in all inertial frames of reference.
  2. The speed of light in a vacuum is the same for all observers regardless of their relative motion or the motion of the source of the light.
- Consequences of Special Relativity include:
  1. Relativity of simultaneity: Two events that are simultaneous in one frame of reference may not be simultaneous in another frame moving relative to the first.
  2. Time dilation: Time intervals between events appear longer for objects in motion relative to an observer at rest.
  3. Mass variation refers to the change in the mass of an object as its speed approaches the speed of light. According to relativity, the mass of an object increases with its velocity, becoming infinite as it reaches the speed of light.
  4. Length contraction: Objects in motion appear shorter in the direction of their motion relative to an observer at rest.
  5. Mass-energy equivalence: Mass and energy are interchangeable $E=mc^2$.
- General Relativity applies to objects under any motion, including acceleration. It explains gravity as a curvature of space-time caused by mass and energy.
- Two postulates of General Relativity are:
  1. The laws of physics may be expressed in equations having the same form in all frames of reference, regardless of their states of motion.
  2. There is no way for an observer in a closed laboratory to distinguish between the effects produced by a gravitational field and those produced by an acceleration of the laboratory itself.
- Consequences of General Relativity include:
  1. Space-time: Space and time are unified into a single four-dimensional fabric.
  2. Gravitational red-shift: Light coming from a strong gravitational field has a longer wavelength (red-shifted).
  3. Black holes: Extremely massive objects whose gravity is so strong that not even light can escape.
  4. Gravitational lensing: Light bends as it travels through curved spacetime, which can distort, diminish or magnify distant objects.
  5. Expansion of the universe: General relativity predicts the universe is expanding, which has been confirmed by astronomical observations.

TEXTBOOK EXERCISE:

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. The General Theory of Relativity was a new way of understanding of: **(2024)**
   (a) The speed of light (b) gravity (c) mass (d) force
2. An object at rest has a mass of 1 kg. What is its mass when it is moving at a speed of 0.9c? **(2025)**
   (a) Infinite (b) 1.2 kg (c) 2.3 kg (d) 1 kg
3. The equivalence of principle in general relativity is:
   (a) The equivalence of inertial and gravitational mass (b) The equivalence of electric and magnetic fields
   (c) The equivalence of space and time (d) The equivalence of matter and energy
4. Which of the following phenomena is NOT predicted by special relativity?
   (a) Time dilation (b) Length contraction
   (c) Gravitational waves (d) Relativistic mass increase
5. What does the equivalence principle state about acceleration and gravity?
   (a) They are completely different forces (b) They are indistinguishable for an observer
   (c) Gravity is stronger than acceleration (d) Acceleration cancels out gravity
6. Light passing near a massive object like a star will bend due to:
   (a) Gravitational lensing (b) refraction (c) reflection (d) diffraction
7. What is the main reason astronomers cannot directly observe black holes?
   (a) They are too small (b) They deflect light
   (c) They are too far away (d) their immense gravity traps light
8. Objects cannot exceed the speed of light because: **(MP:25)**
   (a) Their mass becomes infinite (b) Their length becomes zero
   (c) Time slows down to zero (d) they lose all energy
9. Why is the Galilean transformation not valid at high speeds?
   (a) It cannot handle accelerating frames (b) It violates the constancy of the speed of light
   (c) It only works in flat spacetime (d) It neglects time dilation effects
10. The example of inertial frames in our everyday life is: **(2025)**
    (a) A car accelerating on a highway (b) A train moving smoothly at constant speed
    (c) A person standing on a spinning platform (d) an airplane encountering turbulence

SECTION- B: SHORT ANSWERED QUESTIONS

1. Show that for values of $v << c$, Lorentz transformation reduces to the Galilean transformation.
2. If a particle could move with the velocity of light, how much K.E. would it possess?
3. Explain the difference between Special and General Relativity in simple terms.
4. Differentiate between Inertial Frames of Reference and Non- Inertial Frames of Reference.
5. Why can't any object move at the speed of light?
6. What is the limitation in the Galilean Transformation Equation, and how did Lorentz solve it?
7. Calculate the value of y (Lorentz factor) if the object is moving at the speed of light.

SECTION- C: LONG ANSWERED QUESTIONS

1. State and explain the basic postulates of Einstein's special theory of relativity. Discuss length-contraction, mass variation and time-dilation.

2. Explain the concept of mass-energy equivalence. Derive Einstein's mass-energy relation and demonstrate that 1 atomic mass unit (u) is equivalent to 931 MeV.
3. Discuss the important conclusions derived from General Theory of relativity. What are the experimental observations in favour of these conclusions?
4. How does it principle of relativity differ from the classical Galilean view?
5. Explain the concept of spacetime curvature in general relativity and how it is used to visualize the effects of gravity.
6. Derive the basic equations of the Galilean transformation and explain how they relate the positions and velocities of objects in different inertial frames.
7. Discuss the Lorentz transformation equations in special relativity and how they describe the relationship between space and time coordinates in different inertial frames. Give examples to illustrate their application.

SECTION- D: NUMERICAL

1. A rod 1 meter long is moving along its length with a velocity 0.6c. Calculate its length as it appears to an observer (a) on the earth (b) moving with the rod itself. [Ans: (a) 0.8m, (b) 1m]
2. How fast would a rocket have to go relative to an observer for its length to be contracted to 99% of its length at rest? [Ans: 42. 45 x 10^6m/s]
3. A particle with a proper lifetime of lμs moves through the laboratory at 2.7 x 10^8 m/s. (a) What is its lifetime, as measured by observers in the laboratory? (b) What will be the distance traversed by it before disintegrating? [Ans: (a) 2. 3 x 10^{-6} s, (b) 620m]
4. At what speed is a particle moving if the mass is equal to three times its rest mass? [Ans:$\frac{2\sqrt{2}}{3}c$] **(MP:25)**
5. If 4 kg of a substance is fully converted into energy, how much energy is produced? [Ans: 3.6 x 10^{17}J]
6. Calculate the rest energy of an electron in joules and in electron volts. [Ans: 8. 2 x 10^{-14}J, 0.511MeV]
7. Calculate the K.E. of an electron moving with a velocity of 0.98 times the velocity of light in the laboratory system. [Ans: 3. 3 x 10^{-13} J]
8. At what velocity does the K.E. of a particle equal its rest energy? [Ans: $\frac{\sqrt{3}}{2}$ c]
9. A particle of rest mass m_o moves with speed $c/\sqrt{2}$. Calculate its mass, momentum, total energy and kinetic energy. [Ans: $\sqrt{2}m_o, \sqrt{2}m_oc^2, 0.41m_oc^2$]
10. The nearest star to Earth is Proximal Centauri, 4.31ight- years away. A spaceship with a constant speed of 0.800c relative to the Earth travels toward the star. (a) How much time would elapse on a clock as measured by travelers on the spacecraft? (b) How long does the trip take according to Earth observers? [Ans: (a) 3.22 years; (b) 5.38 years] **(2024)**

ASSIGNEMENT: 24

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. The relativistic mass of 5kh ball observed by observer moving with a speed equal to speed of light is:
   (a) 5 kg (b) 10 kg **(c) infinite** (d) 20 kg **(2022)**
2. This factor is Lorentz factor: **(2021)**
   (a) $1-\frac{v^2}{c^2}$ **(b) $\sqrt{1-\frac{v^2}{c^2}}$** (c) $\sqrt{1-\frac{c^2}{v^2}}$ (d) $\sqrt{\frac{v^2}{c^2}-1}$
3. A frame of reference is called inertial if it is: **(2021, 17, 07)**
   (a) Rotatory (b) accelerated
   (c) vibratory **(d) moving with uniform velocity**
4. This was the first experimental verification of Einstein's mass-energy relation: **(2013)**
   (a) Deutron-induced reaction **(b) Protons-induced reaction**
   (c) Gama induced reaction (d) none of these
5. According to Einstein's special theory of relatively, the mass of a particle moving with the speed of light will become: **(2012)**
   (a) Zero (b) double **(c) infinite** (d) ten times
6. Galilean transformations are applicable to a frame of reference which is: **(2008)**
   (a) stationary (b) moving **(c) inertial** (d) non-inertial.
7. The experimental evidence of Einstein's mass-energy equation is: **(2008)**
   (a) radioactive emission (b) photoelectric and Compton effect
   (c) elastic collision **(d) pair production and annihilation of matter**
8. A frame of reference is called inertial if it is: **(2002E)**
   (a) Rotatory (b) Accelerated
   (c) Moving with uniform velocity (d) Vibratory
9. The relativistic changes in mass, length and time in daily life are not observed because: **(2001)**
   (a) The masses of objects are very large
   (b) The sizes of objects are very large
   (c) The velocity of objects is very small in comparison to the velocity of light
   (d) None of the above

SECTION- B: SHORT QUESTIONS

1. Give the assumptions of special theory of relativity and discuss any one of the results obtained. (2014)
2. What do you understand by the Frame of reference and inertial Frame of Reference? Give one example of each from daily life. Briefly discuss the three basic equations from relativity predicting the relativistic changes. (2005)

NUMERICAL PROBLEMS:

3. Prove that the speed of a moving rod at which its relative length is equal to half of its proper length is $\frac{\sqrt{3}c}{2}$. **(2025)**
4. Find the speed at which mass of a particle will be doubled. (Ans: $\frac{\sqrt{3}}{2}\boldsymbol{c}$) **(2023, 08, 02M)**
5. What will be the velocity and momentum of particle whose rest mass is m_o and whose K.E. (Kinetic Energy) is equal to its rest mass energy. **(2022, 19, 15, 14)**
6. Given m_0c^2= 0.511 MeV. Find the total energy and the kinetic energy of an electron moving with a speed v=0.85 c. (Ans:0.97 MeV, 0.459 MeV) (2012, 02E)

7. What will be the relative velocity and momentum of a particle whose rest mass is m_o and kinetic energy is equal to twice of its rest mass energy. (Ans: $\frac{2\sqrt{2}}{3}c$, $2\sqrt{2}\ m_oc$) (2011)
8. Compare the energy of a photon of wavelength 2×10^{-6} m with the energy of X-Ray photon of wavelength 2×10^{-10} m. (Ans: 10000 times) (2004)
9. An electron has a speed of 10^6 m/s. Find its energy in electron volts. ($m_o = 9.1 \times 10^{-31}$ kg) (Ans: 1.503×10^{-19} kg, 939.37 MeV) (2004)
10. If a neutron is converted entirely into energy, how much energy is produced? Express your answer in joule and electron-volt. Given: $m_n = 1.67 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m/s. (2003)

SECTION- C: DETAILED QUESTIONS:

11. Write the postulates of special theory of relativity. Explain any two of its results. **(2025)**
12. What are inertial and non-inertial frames of references? Write down the postulates of special theory of relativity and briefly discuss its consequences.
13. State the basic postulates of the Special Theory of Relativity? Explain the four important results of Einstein's Theory of Relativity. **(2021)**
14. Write the principle of relativity and the two postulates of special theory of relativity. **(2018)**
15. What are inertial and non-inertial frames of references? Also give the consequences of the special theory of relativity. (2017)
16. What are the postulates of the Special Theory of Relativity? Explain any three results of Einstein's Theory of Relativity. (2006, 99)
17. Give the postulates of the Special Theory of Relativity and discuss the important results obtained by Einstein. (2002E)
18. Give the postulates and explain the consequences of Special Theory of Relativity. (2002M)